

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 3 Study of Pembrolizumab in Combination With Pemetrexed/Platinum (Carboplatin or Cisplatin) Followed by Pembrolizumab and Maintenance Olaparib vs Maintenance Pemetrexed in the First-Line Treatment of Participants With Metastatic Nonsquamous Non-Small-Cell Lung Cancer (NSCLC)
Short Title:	Study of Pembrolizumab With Maintenance Olaparib or Maintenance Pemetrexed in First-line (1L) Metastatic Nonsquamous Non-Small-Cell Lung Cancer (NSCLC) (MK-7339-006, KEYLYNK-006)
Protocol Number:	681501866234303631
NCT Number:	NCT03976323
Sponsor Name:	Merck
Investigational Product:	cisplatin
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Progression-free Survival (PFS): PFS was defined as the time from randomization to the first documented progressive disease (PD) per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) based on blinded independent central review (BICR) or death due to any cause, whichever occurs first. PD was defined as ≥20% increase in the sum of diameters of target lesions. In addition to the relative increase of 20%, the sum must also have demonstrated an absolute increase of ≥5 mm. The appearance of one or more new lesions was also considered PD. PFS was analyzed by the non-parametric Kaplan-Meier (K-M) method. The protocol specified final analysis of PFS is presented here for the first pembrolizumab course in the Maintenance Phase. Per protocol, analysis for this outcome measure was not planned or conducted in the Induction Phase. 2. Overall Survival (OS): OS was defined as the time from randomization to death due to any cause. OS was analyzed by the non-parametric K-M method. Participants without documented death at the time of analyses were censored at the date of last known to be alive. The protocol specified final analysis of OS is presented here for the first pembrolizumab course in the Maintenance Phase. Per protocol, analysis for this outcome measure was not planned or conducted in the Induction Phase.
Secondary Obj.:	1. Number of Participants Who Experience One or More Adverse Events (AEs) 2. Number of Participants Who Discontinue Study Treatment Due to an AE 3. Change From Baseline in European Organization for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire-Core 30 (QLQ-C30) Global Health Status /

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Quality of Life (Items 29 and 30) Scale Score

2.2 Background & Rationale

The current study will compare pembrolizumab (MK-3475) plus maintenance olaparib, versus (vs) pembrolizumab plus maintenance pemetrexed for the treatment of non-squamous NSCLC. The study's 2 primary hypotheses are: 1. Pembrolizumab plus maintenance olaparib is superior to pembrolizumab plus maintenance pemetrexed with respect to progression-free survival (PFS) per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) by blinded independent clinical review (BICR) and 2. Pembrolizumab plus maintenance olaparib is superior to pembrolizumab plus maintenance pemetrexed with respect to overall survival (OS).

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: ACTIVE_NOT_RECRUITING

Target N: 1003

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. Have a histologically or cytologically confirmed diagnosis nonsquamous NSCLC.
2. Have stage IV nonsquamous NSCLC.
3. Have confirmation that epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase (ALK), or Proto-oncogene tyrosine-protein kinase (ROS1)-directed therapy is not indicated.
4. Have measurable disease based on RECIST 1.1.
5. Have provided archival tumor tissue sample or newly obtained core or incisional biopsy of a tumor lesion not previously irradiated.

Note: Adequacy of biopsy specimen for the above analyses must be confirmed by the central laboratory before the participant can start the induction phase. Submission of another tumor specimen may be required prior to enrolling the participant, if adequate tumor tissue was not provided the first time.

6. Have a life expectancy of at least 3 months.
7. Have a performance status of 0 or 1 on the Eastern Cooperative Oncology Group (ECOG) Performance Status assessed within 7 days prior to the administration of study intervention.
8. Have not received prior systemic treatment for their advanced/metastatic NSCLC.
9. Have adequate organ function.
10. Male and female participants who are not pregnant and of childbearing potential must follow contraceptive guidance during the treatment period and for 180 days afterwards.
11. Male participants must refrain from donating sperm, and female participants must refrain from donating eggs to others or freeze/store for her own use during the treatment period and for 180 days afterwards.

4.2 Exclusion Criteria

1. Has predominantly squamous cell histology NSCLC.
2. Has a known additional malignancy that is progressing or has progressed within the past 3 years requiring active treatment.

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3. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis.
4. Has a severe hypersensitivity (?Grade 3) to pembrolizumab and/or any of its excipients.
5. Has a known hypersensitivity to any components or excipients of cisplatin, carboplatin, pemetrexed, or olaparib.
6. Has an active autoimmune disease that has required systemic treatment in past 2 years.
7. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy.
8. Has a known history of human immunodeficiency virus (HIV) infection, a known history of hepatitis B infection, or known active hepatitis C virus infection.
9. Has interstitial lung disease, or history of pneumonitis requiring systemic steroids for treatment.
10. Has received prior therapy with olaparib or with any other polyadenosine 5' diphosphoribose (polyADP ribose) polymerization (PARP) inhibitor.
11. Has received prior therapy with an agent directed to programmed cell death ligand 1 (PD-L1), anti PD-L2, or directed to a stimulatory or co-inhibitory T-cell receptor (e.g., cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), OX-40, CD137).
12. Has myelodysplastic syndrome (MDS)/acute myeloid leukemia (AML) or with features suggestive of MDS/AML.
13. Has history of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years.
14. Has completed palliative radiotherapy within 7 days of the first dose. Participants must have recovered from all radiation-related toxicities and not require corticosteroids.

11. Study Identifiers & Governance

Primary ID:	681501866234303631
Registry Number:	NCT03976323
Sponsor:	Merck
Study Title:	Study of Pembrolizumab With Maintenance Olaparib or Maintenance Pemetrexed in First-line (1L) Metastatic Nonsquamous Non-Small-Cell Lung Cancer (NSCLC) (MK-7339-006, KEYLYNK-006)

15. Sign-off & Approval

Principal Investigator: _____	Date: _____
Clinical Operations Lead: _____	Date: _____
Lead Statistician: _____	Date: _____